

NON-LINEAR NON-PLANAR OSCILLATIONS OF A CANTILEVER BEAM UNDER LATERAL BASE EXCITATIONS

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Abstract—The non-planar responses of a cantilevered beam subject to lateral harmonic base-excitation are investigated using two non-linear coupled integrodifferential equations of motion. The equations contain cubic non-linearities due to curvature and inertia. Two uniform beams with rectangular cross sections are considered: one has an aspect ratio near unity, and the other has an aspect ratio near 6.27. A combination of the Galerkin procedure and the method of multiple scales is used to construct a first-order uniform expansion for the case of a one-to-one internal resonance and a primary resonance. The results show that the non-linear geometric terms are important for the responses of low-frequency modes because they produce hardening spring effects. On the other hand, the non-linear inertia terms dominate the responses of high-frequency modes. We also obtain quantitative results for non-planar motions and investigate their dynamic behavior. For different range of parameters, the non-planar motions can be steady whirling motions, whirling motions of the beating type, or chaotic motions. Furthermore, we investigate the effects of damping.

1. INTRODUCTION

The dynamic behavior of a long, slender beam is the subject of interest in connection with helicopter rotor blades, manipulator arms, spacecraft antennas, flexible satellites, and other systems that perform large and/or complex motions.

Linear formulation of beam problems shows that the motions in the two principal planes are independent; consequently, forced motion in one principal plane is always stable in that plane. But when the vibration amplitude is large, various beam effects (e.g. shear deformation, warping, rotary inertia, gravity, etc.) become significant, and several non-linearities (due to inertia, curvature, mid-plane stretching, lateral static loads, material, damping) come into play. And some of the beam effects will introduce extra non-linear terms in the equations of motion. To derive the equations of motion including all of the beam effects and non-linearities is almost impossible. In the existing literature, one often finds that a particular effect is introduced in an *ad hoc* fashion and all other effects are neglected based on the authors' judgement.

For a slender rod subject to lateral harmonic excitation, Haight and King [1] investigated the effect of non-linear inertia terms. They identified unstable regions in the planar response curves, which correspond to parametrically excited non-planar motions, without obtaining quantitative results about the resulting non-planar motion. From experimental work, they concluded that the non-planar motions are steady whirling motions. They also stated that damping could cause the disappearance of the planar instability phenomenon. Hyer [2] used Haight and King's equations to analyse quantitatively the whirling motion of the same problem. He concluded that there were no unstable whirling motions. But by including non-linear curvature terms in the equations of motion for the same problem, Crespo da Silva [3] found that for some range of the excitation frequency, no stable planar or non-planar responses exist.

For large amplitude whirling motions of a simply supported, axially constrained beam, Ho, Scott and Easley [4] showed that the non-linear terms due to stretching are cubic and cause whirling motions. They found that, in some parameter range, both planar and non-planar motions could occur were possible. Nayfeh *et al.* [5-9] investigated the effects of stretching, transverse shear, rotary inertia and material non-linearity. They analysed primary and secondary resonances in the presence and absence of modal coupling and demonstrated the occurrence of period-doubling bifurcations and chaos. Hodges and Dowell [10]

used an exact coordinate transformation and an ordering scheme to develop a comprehensive set of equations with quadratic non-linearities to describe the dynamics of beams. They also identified several non-linear terms. Dowell, Traybar and Hodges [11] devised a simple, non-rotating beam with a tip weight to evaluate the theory of Hodges and Dowell [10]. The results show that there are systematic differences between theory and experiment when tip deflections become large. Rosen and Friedmann [12] derived a more accurate set of equations than those of Hodges and Dowell [10] by including some non-linear terms with order higher than two. Their numerical results are in good agreement with the experimental data of [11]. Tso [13] studied torsional vibrations due to shortening effects of an elastic cantilever of rectangular cross section under dynamic axial loading. Sato, Saito and Otomi [14] treated parametric vibrations of a simply supported horizontal beam, carrying a concentrated mass under the influence of gravity.

Although Bolotin [15] and Moody [16] showed that non-linear inertia effects are far more significant than non-linear elasticity effects on the vibrations of thin rods, Crespo da Silva and Glynn [17, 18] showed that the generally neglected non-linear terms arising from the curvature are the same order as the non-linear terms due to inertia. An experimental study by Evan-Iwanowski, Sanford and Kehagioglou [19] also shows that non-linear geometric terms have definite effects on the non-stationary parametric response of a straight column. Crespo da Silva and Hodges [20, 21] derived the differential equations of motion for a rotating beam with the objective of retaining all possible contributions due to higher-order non-linearities and investigated the influence of these terms on the motion of a helicopter rotor blade. For prismatic, uniform, non-rotating beams, the equations of motion of [20] and [21] reduce to those of [17]. Also, Burton and Kolowith [22] and Haddow and Hasan [23] observed chaotic motions, non-planar motions, and in-plane modal interactions in their experimental work on a parametrically excited, vertically mounted beam.

Here we use the equations of motion derived by Crespo da Silva and Glynn [17] to analyse the response of a slender cantilever beam to a transverse harmonic base excitation. We use a combination of the method of multiple scales [24] and the Galerkin procedure to determine ranges of the driving frequency which result in unstable planar motions and describe quantitatively the non-planar motions. We determine some of the main characteristics of the non-planar motions and investigate the possible bifurcations of the response amplitudes when the excitation frequency is varied. We also demonstrate the existence of Hopf bifurcations and the non-existence of periodic motions for certain excitation parameters.

2. PROBLEM FORMULATION

In order to use the equations derived by Crespo da Silva and Glynn [17] to analyse the forced vibration behavior of a fixed-free beam as shown in Fig. 1 and to simplify the problem, we use the following notations: x , y and z are the inertial coordinates; t is the time; s is the undeformed arc length (material coordinate); ξ , η and ζ are the principal axes of the beam's cross section at position s ; D_ξ , D_η and D_ζ are the principal stiffnesses; $\beta_y \equiv D_\zeta/D_\eta$ and $\beta_z \equiv D_\xi/D_\eta$; $u(s, t)$, $v(s, t)$ and $w(s, t)$ are the components of the displacement of the centroid at an arbitrary location s along the inertial axes x , y and z , respectively; and g denotes the acceleration due to gravity. And we make the following assumptions: (a) the cross-section dimensions b and h and material properties of the beam are uniform, (b) the distributed torsional moments of inertia of the beam are negligible (this implies that only beams for which the torsional frequency is higher than the frequencies of flexural vibrations can be used in the following analysis), (c) the dissipation of energy due to internal friction, air resistance, and relative motion between the beam and its support system can be modeled by a viscous damper having the coefficient c and the virtual work $-c(\dot{v}\delta v + \dot{w}\delta w)$, and (d) $w(s, t) = \bar{w}(s, t) + F \cos \Omega t$, where $\bar{w}$ denotes the displacement in the z -direction, with respect to the base, and F and Ω are the constant amplitude and frequency of the base motion.

All variables are non-dimensionalized using the constrained length of the beam L and the characteristic time $L^2 \sqrt{m/D_\eta}$, where m is the mass per unit length. Then the equations of

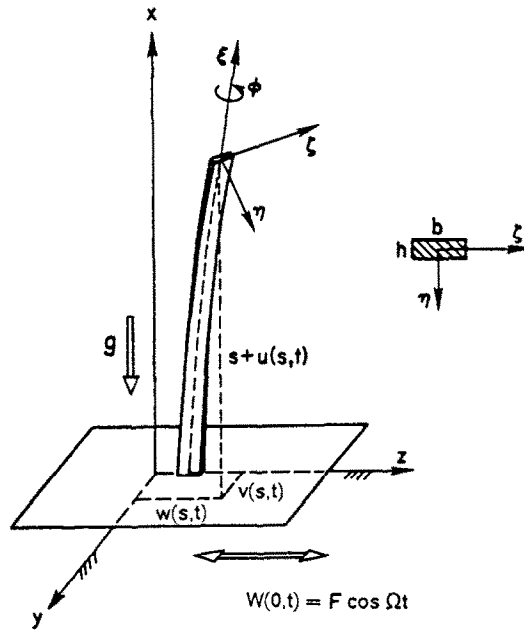


Fig. 1. Coordinate systems: x - y - z = inertial reference frame; ξ - η - ζ = principal axes of the beam's cross section at position s , which are fixed on the cross section.

motion can be written in non-dimensional form as [17, 18]

$$\begin{aligned} \ddot{v} + c\dot{v} + \beta_y v^{iv} = & (1 - \beta_y) \left[w'' \int_1^s v'' w'' ds - w''' \int_0^s v'' w' ds \right]' \\ & - \frac{(1 - \beta_y)^2}{\beta_y} \left[w'' \int_0^s \int_1^s v'' w'' ds ds \right]'' - \beta_y [v'(v'v'' + w'w'')]'' \\ & - \frac{1}{2} \left\{ v' \int_1^s \left[\int_0^s (v'^2 + w'^2) ds \right]'' ds \right\}' + [v''(s-1) + v'] \frac{L^3}{D_\eta} mg \end{aligned} \quad (1a)$$

$$\begin{aligned} \ddot{w} + c\dot{w} + w^{iv} = & -(1 - \beta_y) \left[v'' \int_1^s v'' w'' ds - v''' \int_0^s w'' v' ds \right]' \\ & - \frac{(1 - \beta_y)^2}{\beta_y} \left[v'' \int_0^s \int_1^s v'' w'' ds ds \right]'' - [w'(w'w'' + v'v'')]'' \\ & - \frac{1}{2} \left\{ w' \int_1^s \left[\int_0^s (v'^2 + w'^2) ds \right]'' ds \right\}' + [w''(s-1) + w'] \frac{L^3}{D_\eta} mg \\ & + F\Omega^2 \cos \Omega t + cF\Omega \sin \Omega t. \end{aligned} \quad (1b)$$

Boundary conditions:

$$v = w = v' = w' = 0 \quad \text{at } s = 0 \quad (2a)$$

$$v''' = w''' = v'' = w'' = 0 \quad \text{at } s = 1. \quad (2b)$$

The forcing terms and the term due to gravity are obtained by using d'Alembert principle. In equations (1a) and (1b), the terms in the first and second square brackets represent the coupling between torsion and lateral vibrations, the second term in the third square bracket represents the coupling between the lateral vibrations in the y - and z -directions, those in the curly bracket are the non-linear inertia terms, the terms in the last square bracket are due to the gravitational load and the use of the inextensibility constraint, and $F\Omega^2 \cos \Omega t$ is the forcing function due to the base motion. Apparently, all the non-linear terms are cubic. And if $\beta_y = 1$, it is obvious that there is no coupling between the torsion and lateral vibrations.

The terms $[w''(s-1) + w'] \frac{L^3}{D_n} mg$ and $[v''(s-1) + v'] \frac{L^3}{D_n} mg$ are found to be third-order terms, as shown later.

3. METHOD OF SOLUTION

To analyse the coupled non-linear equations (1a) and (1b), we employ the method of multiple scales [24]. Toward this end a small parameter ε is introduced as a bookkeeping device in the following perturbation analysis. Because all non-linear terms are cubic, we seek a first-order uniform expansion, and we consider weakly non-linear vibrations. Thus, we assume that an approximation to $v(s, t)$ and $w(s, t)$ can be expressed as

$$v(s, T_0, T_2; \varepsilon) = \varepsilon v_1(s, T_0, T_2, \dots) + \varepsilon^3 v_3(s, T_0, T_2, \dots) + \dots \quad (3)$$

$$w(s, T_0, T_2; \varepsilon) = \varepsilon w_1(s, T_0, T_2, \dots) + \varepsilon^3 w_3(s, T_0, T_2, \dots) + \dots \quad (4)$$

Here, $T_0 = t$ is a fast scale, characterizing motions with Ω and the natural frequencies ω_{mn} , and $T_2 = \varepsilon^2 t$ is a slow scale, characterizing the modulations of the amplitudes and phases. Moreover, we let $c = \varepsilon^2 \mu$, $F = \varepsilon^3 f$, and $\beta_y = 1 + \delta_0 + \varepsilon^2 \delta_2$. Substituting equations (3) and (4) into equations (1a) and (1b) and equating coefficients of like powers of ε , we obtain

Order ε :

$$D_0^2 v_1 + (1 + \delta_0) v_1^{iv} = 0 \quad (5a)$$

$$D_0^2 w_1 + w_1^{iv} = 0 \quad (5b)$$

$$v_1 = w_1 = v_1' = w_1' = 0 \quad \text{at } s = 0 \quad (6a)$$

$$v_1'' = w_1'' = v_1''' = w_1''' = 0 \quad \text{at } s = 1 \quad (6b)$$

Order ε^3 :

$$\begin{aligned} D_0^2 v_3 + (1 + \delta_0) v_3^{iv} = & -2D_0 D_2 v_1 - \mu D_0 v_1 - \frac{(1 - \beta_y)^2}{\beta_y} \left[w_1'' \int_0^s \int_1^s v_1' w_1' ds ds \right]'' \\ & - \delta_2 v_1^{iv} + (1 - \beta_y) \left[w_1'' \int_1^s v_1'' w_1'' ds - w_1''' \int_0^s v_1'' w_1' ds \right]' \\ & - \beta_y [v_1'(v_1' v_1'' + w_1' w_1'')] - \left\{ \frac{v_1'}{2} \int_1^s \left[\int_0^s (v_1'^2 + w_1'^2) ds \right]'' ds \right\}' \\ & + [v_1''(s-1) + v_1'] g_0 \end{aligned} \quad (7a)$$

$$\begin{aligned} D_0^2 w_3 + w_3^{iv} = & -2D_0 D_2 w_1 - \mu D_0 w_1 - (1 - \beta_y) \left[v_1'' \int_1^s v_1'' w_1'' ds - v_1''' \int_0^s w_1'' v_1' ds \right]' \\ & - \frac{(1 - \beta_y)^2}{\beta_y} \left[v_1'' \int_0^s \int_1^s v_1'' w_1'' ds ds \right]'' - \left\{ \frac{w_1'}{2} \int_1^s \left[\int_0^s (v_1'^2 + w_1'^2) ds \right]'' ds \right\}' \\ & - [w_1'(v_1' v_1'' + w_1' w_1'')] + [w_1''(s-1) + w_1'] g_0 + b\Omega^2 \cos(\Omega T_0) \end{aligned} \quad (7b)$$

$$v_3 = w_3 = v_3' = w_3' = 0 \quad \text{at } s = 0 \quad (8a)$$

$$v_3'' = w_3'' = v_3''' = w_3''' = 0 \quad \text{at } s = 1 \quad (8b)$$

where $D_n \equiv \partial/\partial T_n$ and $g_0 \equiv L^3 mg/(D_n \varepsilon^2)$.

The general solution of (5) and (6) consists of two infinite sequences of modes corresponding to two infinite sequences of frequencies. Any mode that is not directly excited or indirectly excited will decay to zero with time due to the presence of damping [5]. In this paper, we consider the primary-resonant excitation of one of the modes in the z direction, say the n th mode, and which in turn excites one of the modes in the y direction, say the m th mode, through a one-to-one internal or autoparametric resonance. Thus, the former mode is directly excited by the primary resonance and the latter mode is indirectly excited through the autoparametric resonance. We assume that none of the other modes in the y and z directions is involved in any autoparametric resonance with either the m th mode in the

y direction or the n th mode in the z direction. Consequently, the non-decaying solutions of (5) and (6) can be expressed as

$$\begin{aligned} v_1(s, T_0, T_2, \dots) &= \Phi_m(s) A_1(T_2) e^{i\omega_{1m} T_0} + \text{cc} \\ w_1(s, T_0, T_2, \dots) &= \Phi_n(s) A_2(T_2) e^{i\omega_{2n} T_0} + \text{cc} \end{aligned}$$

where cc stands for the complex conjugate of the preceding terms,

$$\omega_{1m} = z_m^2 \sqrt{1 + \delta_0}, \quad \omega_{2n} = z_n^2,$$

$$\Phi_i(s) = \cosh z_i s - \cos z_i s + \frac{\cos z_i + \cosh z_i}{\sin z_i + \sinh z_i} (\sin z_i s - \sinh z_i s).$$

Here, z_i is the root of $1 + \cos z \cosh z = 0$. Its lowest three roots are 1.8751, 4.6941, and 7.8548. The eigenfunctions $\Phi_i(s)$ satisfy

$$\int_0^1 \Phi_i^2(s) ds = 1.$$

We note that ω_{2n} is the linear natural frequency of the n th mode of vibration in the z direction and ω_{1m} is approximately equal to the linear natural frequency of the m th mode of vibration in the y direction. The latter actual frequency is $z_m^2 \sqrt{1 + \delta_0 + \varepsilon^2 \delta_2}$.

To determine the equations describing the functions $A_i(T_2)$, we use the Galerkin procedure to reduce (7) and (8) into ordinary-differential equations. To this end, we let

$$v_{1t}(T_0, T_2) = A_1(T_2) e^{i\omega_{1m} T_0} + \text{cc} \quad (9a)$$

$$v_{3t}(T_0, T_2) = \int_0^1 \Phi_m(s) v_3(s, T_0, T_2) ds \quad (9b)$$

$$w_{1t}(T_0, T_2) = A_2(T_2) e^{i\omega_{2n} T_0} + \text{cc} \quad (9c)$$

$$w_{3t}(T_0, T_2) = \int_0^1 \Phi_n(s) w_3(s, T_0, T_2) ds. \quad (9d)$$

Multiplying (7a) and (7b) by $\Phi_m(s)$ and $\Phi_n(s)$, respectively, and integrating over the domain of s , we obtain

$$\begin{aligned} D_0^2 v_{3t} + \omega_{1m}^2 v_{3t} &= -2D_0 D_2 v_{1t} - \mu D_0 v_{1t} - \delta_2 z^4 v_{1t} \\ &+ (1 - \beta_y) \alpha_1 v_{1t} w_{1t}^2 - \frac{(1 - \beta_y)^2}{\beta_y} \alpha_2 v_{1t} w_{1t}^2 - \beta_y \alpha_3 v_{1t} w_{1t}^2 - \beta_y \alpha_4 v_{1t}^3 \\ &- \frac{1}{2} \alpha_5 v_{1t} D_0^2 v_{1t}^2 - \frac{1}{2} \alpha_6 v_{1t} D_0^2 w_{1t}^2 + \alpha_7 g_0 v_{1t} \end{aligned} \quad (10)$$

$$\begin{aligned} D_0^2 w_{3t} + \omega_{2n}^2 w_{3t} &= -2D_0 D_2 w_{1t} - \mu D_0 w_{1t} \\ &- (1 - \beta_y) \beta_1 v_{1t}^2 w_{1t} - \frac{(1 - \beta_y)^2}{\beta_y} \beta_2 v_{1t}^2 w_{1t} - \beta_3 v_{1t}^2 w_{1t} - \beta_4 w_{1t}^3 \\ &- \frac{1}{2} \beta_5 w_{1t} D_0^2 w_{1t}^2 - \frac{1}{2} \beta_6 w_{1t} D_0^2 v_{1t}^2 + \beta_7 g_0 w_{1t} + \beta_8 f \Omega^2 \cos \Omega T_0 \end{aligned} \quad (11)$$

where the α_i and β_i are defined in Appendix A.

To investigate the system response under a primary-resonant excitation, we introduce the detuning parameter $\hat{\sigma}$ defined by

$$\Omega = \omega_{2n}(1 + \varepsilon^2 \hat{\sigma}) \quad (12)$$

and we let $\omega_{1m} = \omega_{2n}$, which represents a one-to-one internal resonance. We assume that none of the other modes is involved in an autoparametric resonance with these modes. Substituting (9a) and (9c) into (10) and (11) and eliminating the terms that produce secular

terms, we obtain the solvability conditions

$$-i(2\omega_{1m}A'_1 + \mu\omega_{1m}A_1) - \delta_2 z^4 A_1 - \left[\delta_0 \alpha_1 + \frac{\delta_0^2}{\beta_\gamma} \alpha_2 + (1 + \delta_0) \alpha_3 \right] (2A_1 A_2 \bar{A}_2 + \bar{A}_1 A_2^2) \\ - 3(1 + \delta_0) \alpha_4 A_1^2 \bar{A}_1 + 2\alpha_5 \omega_{1m}^2 \bar{A}_1 A_1^2 + 2\alpha_6 \omega_{2n}^2 \bar{A}_1 A_2^2 + \alpha_7 g_0 A_1 = 0 \quad (13a)$$

$$-i(2\omega_{2n}A'_2 + \mu\omega_{2n}A_2) + \left(\delta_0 \beta_1 - \frac{\delta_0^2}{\beta_\gamma} \beta_2 - \beta_3 \right) (2A_2 A_1 \bar{A}_1 + \bar{A}_2 A_1^2) - 3\beta_4 A_2^2 \bar{A}_2 \\ + 2\beta_5 \omega_{2n}^2 \bar{A}_2 A_2^2 + 2\beta_6 \omega_{1m}^2 \bar{A}_2 A_1^2 + \beta_7 g_0 A_2 + \frac{1}{2} \beta_8 f \omega_{2n}^2 e^{i\omega_{2n} \hat{\sigma} T_2} = 0. \quad (13b)$$

Substituting the polar form

$$A_1 = \frac{1}{2} a_1(T_2) e^{i\theta_1(T_2)} \quad \text{and} \quad A_2 = \frac{1}{2} a_2(T_2) e^{i\theta_2(T_2)}$$

into (13) and separating real and imaginary parts, we obtain the modulation equations

$$2\omega_{1m}a'_1 + [R_1 + R_2 a_2^2 \sin 2(\gamma_1 - \gamma_2)] a_1 = 0 \quad (14)$$

$$[2\omega_{1m}\gamma'_1 - R_3 - R_4 a_1^2 + R_5 a_2^2 + R_2 a_2^2 \cos 2(\gamma_1 - \gamma_2)] a_1 = 0 \quad (15)$$

$$2\omega_{2n}a'_2 + [E_1 + E_2 a_1^2 \sin 2(\gamma_1 - \gamma_2)] a_2 - \beta_8 f \omega_{2n}^2 \sin \gamma_2 = 0 \quad (16)$$

$$[2\omega_{2n}\gamma'_2 - E_3 - E_4 a_2^2 - E_5 a_1^2 - E_2 a_1^2 \cos 2(\gamma_1 - \gamma_2)] a_2 - \beta_8 f \omega_{2n}^2 \cos \gamma_2 = 0 \quad (17)$$

where R_i and E_i are defined in Appendix B and

$$\gamma_1 = \omega_{2n} \hat{\sigma} T_2 - \theta_1, \quad \gamma_2 = \omega_{2n} \hat{\sigma} T_2 - \theta_2. \quad (18)$$

Periodic solutions of the system correspond to the fixed points (i.e. constant solutions) of (14)–(18), which in turn correspond to $a'_1 = a'_2 = \gamma'_1 = \gamma'_2 = 0$. It follows from (18) that $\theta'_1 = \omega_{2n} \hat{\sigma}$ and $\theta'_2 = \omega_{2n} \hat{\sigma}$. Thus, equations (14)–(18) reduce to a set of algebraic equations that can be solved numerically using, for example, the false position method [25].

And to the first-order approximation the response of the beam is given by

$$v(s, t) = \varepsilon \Phi_m(s) a_1(\varepsilon^2 t) \cos(\Omega t - \gamma_1) + \dots \quad (19)$$

$$w(s, t) = \varepsilon \Phi_n(s) a_2(\varepsilon^2 t) \cos(\Omega t - \gamma_2) + \dots \quad (20)$$

To determine the stability of the fixed points, we let

$$p_i = a_i \cos \gamma_i \quad \text{and} \quad q_i = a_i \sin \gamma_i. \quad (21)$$

Using (14)–(18), and (24), we find that

$$-2\omega_{1m}p'_1 = R_3 q_1 + R_4(p_1^2 + q_1^2)q_1 - R_5(p_2^2 + q_2^2)q_1 + R_1 p_1 \\ + R_2[-2q_2 p_2 p_1 + q_1(p_2^2 - q_2^2)] \quad (22)$$

$$-2\omega_{1m}q'_1 = -R_3 p_1 - R_4(p_1^2 + q_1^2)p_1 + R_5(p_2^2 + q_2^2)p_1 + R_1 q_1 \\ + R_2[(p_2^2 - q_2^2)p_1 + 2q_1 p_2 q_2] \quad (23)$$

$$-2\omega_{2n}p'_2 = E_3 q_2 + E_4(p_2^2 + q_2^2)q_2 + E_5(p_1^2 + q_1^2)q_2 + E_1 p_2 \\ + E_2[q_2(q_1^2 - p_1^2) + 2p_2 p_1 q_1] \quad (24)$$

$$-2\omega_{2n}q'_2 = -E_3 p_2 - E_4(p_2^2 + q_2^2)p_2 - E_5(p_1^2 + q_1^2)p_2 + E_1 q_2 \\ - E_2[p_2(p_1^2 - q_1^2) + 2q_2 p_1 q_1] - \beta_8 f \omega_{2n}^2. \quad (25)$$

As equations (22)–(25) are first-order autonomous differential equations, the stability of a particular fixed point with respect to an infinitesimal disturbance proportional to $\exp(\lambda T_2)$ is determined by the eigenvalues of the Jacobian matrix of their right-hand sides. A given fixed point is stable if and only if the real parts of all eigenvalues are less than or equal to zero. From the expression of R_3 and E_3 in Appendix B, we know that owing to the effect of gravity, the natural frequencies of $v(s, t)$ and $w(s, t)$ are decreased by

$\alpha_7 g_0 / (200 \omega_{1m} \omega_{2n})$ % and $\beta_7 g_0 / (200 \omega_{2n}^2)$ %, respectively, which are very small. For example, by checking the dimensions and the material properties of the beams used in [1, 22, 23], we find that the maximum value is 0.96 %, which occurs for the first-mode of Haight and King's beam. Thus, we put

$$\sigma \equiv \delta + \frac{\alpha_7 g_0}{2 \omega_{1m} \omega_{2n}} \quad \text{for } v(s, t) \quad \text{and} \quad \sigma \equiv \delta + \frac{\beta_7 g_0}{2 \omega_{2n}^2} \quad \text{for } w(s, t)$$

in the following analysis.

4. NUMERICAL RESULTS

We investigate the responses of two different beams to lateral harmonic excitations. One beam has an almost square cross section, which implies that the natural frequency of a given mode of $v(s, t)$ is close to the natural frequency of the same mode of $w(s, t)$. The other beam has a cross section with an aspect ratio b/h close to 6.2673; in this case, the natural frequency of the second mode of $v(s, t)$ is close to that of the first mode of $w(s, t)$. For all the following cases, we assume that the beam is uniform, the material is homogeneous and isotropic and obeys Hooke's law, Poisson's ratio $\nu = 0.3$, and the cross section is rectangular with width b and height h . From material mechanics and the theory of elasticity, we know that

$$\beta_y = D_\zeta / D_\eta = (h/b)^2 \simeq 1 + \delta_0 \quad \text{and} \quad \beta_y = D_\xi / D_\eta = 6k_1 \frac{(h/b)^2}{(1 + \nu)} \quad (26)$$

where $D_\xi = k_1 b h^3 E / (2 + 2\nu)$, k_1 can be calculated using equations (168) and (171) of ref. [26], and E is Young's modulus.

It follows from (13) that the effective non-linear spring coefficients α_e and β_e in the y and z directions are

$$\alpha_e = (1 + \delta_0) \alpha_4 - \frac{2}{3} \alpha_5 \omega_{1m}^2 \quad \text{and} \quad \beta_e = \beta_4 - \frac{2}{3} \beta_5 \omega_{2n}^2. \quad (27)$$

Since $\alpha_4, \alpha_5, \beta_4$ and β_5 are positive according to Appendix A, the non-linear geometric terms $(1 + \delta_0) \alpha_4$ and β_4 are of the hardening type, whereas the non-linear inertia terms $-\frac{2}{3} \alpha_5 \omega_{1m}^2$ and $-\frac{2}{3} \beta_5 \omega_{2n}^2$ are of the softening type. Hence, the overall type of the effective non-linearity in the $y(z)$ direction depends on the relative magnitudes of $\alpha_4(\beta_4)$ and $\alpha_5(\beta_5)$ and the mode shape.

It follows from (15) and (16) that $v(s, t)$ can only be excited indirectly through the autoparametric resonance and that the only directly excited motion is a planar motion in the z -direction. Unless specified, we investigate the stability by assuming disturbances in both the y - and z -directions. We note that $\sigma = 0$ and $\sigma = \delta_2 / 2(1 + \delta_0)$ correspond to the pure linear resonant points for $w(s, t)$ and $v(s, t)$, respectively. Since $F (= \epsilon^3 f L)$ is a third-order quantity and $v(s, t)$ and $w(s, t)$ are first-order quantities, we note that, under primary resonance, a small base motion can result in large responses.

Case 1: near square cross section

In this case, $\delta_0 = 0.0$ and $\beta_y = 0.6489$. We let $\mu = 0.04$. For a beam with square cross-section, the planar response curves of the first mode with different amplitudes of the base excitation are shown in Fig. 2. We note that when the amplitude of the base motion f is small, the response amplitude is small, single-valued, and stable, which corresponds to the linear solution. The phase angles are not constant. If the input frequency is away from the perfect resonant point $\sigma = 0$, γ_2 is close to 0° for $\sigma < 0$ and 180° for $\sigma > 0$ as predicted by linear theory. As f increases, the response curves are bent to the right, implying that the non-linear geometric terms dominate the response and produce an effective hardening spring effect. It is well known that, if there are only in-plane disturbances, the branch AB is unstable but the branches BC and DE are stable [5]. Now owing to out-of-plane disturbances (i.e. disturbances perpendicular to the direction of motion), the branches BC and DE are unstable. Hence, the notion that non-linearities are only important for large-excitation levels is not true. Figure 3 shows the frequency-response curves of the second mode for

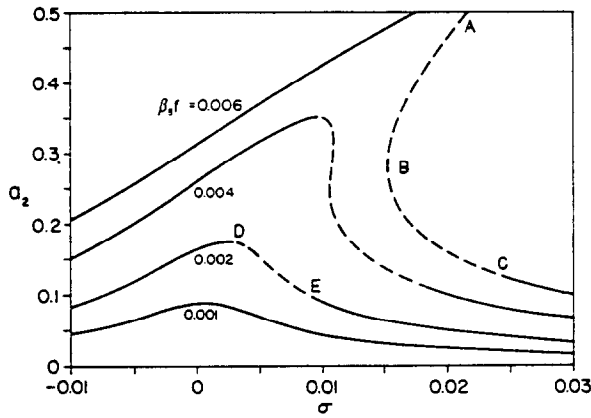


Fig. 2. Planar response curves of the first mode for a beam with an aspect ratio $b/h \approx 1.0$: mode (1, 1), $\omega_{11} = \omega_{21}$, $\delta_0 = 0.0$, $\delta_2 = 0.0$, $\mu = 0.04$, $\beta_y = 0.6489$; (—) stable, (---) unstable with at least one eigenvalue being positive.

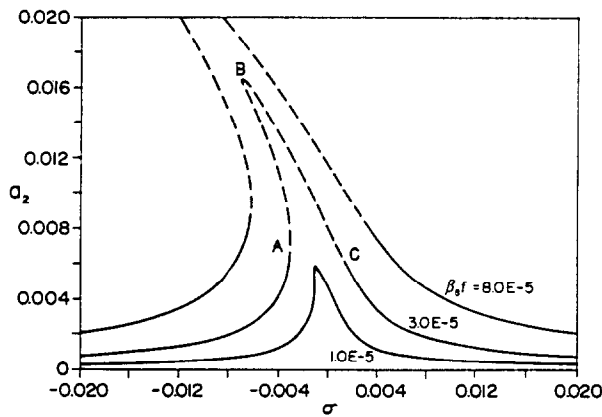


Fig. 3. Planar response curves of the second mode for a beam with an aspect ratio $b/h \approx 1.0$: mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.0$, $\mu = 0.04$, $\beta_y = 0.6489$; (—) stable, (---) unstable with at least one eigenvalue being positive.

different values of f . Apparently, the non-linear inertia terms dominate the response and produce an effective softening spring effect. Again the branch BC is unstable due to out-of-plane disturbances, and the branch AB is unstable due to in-plane disturbances. Furthermore, away from the perfect resonant point $\sigma = 0$, the phase angles of the left and right branches are close to 0° and 180° , respectively. As the response curves of high-frequency modes are similar to those of the second mode, we concentrate on the first two modes only.

When we decrease the detuning δ_2 from 0.0 to -0.01 , the unstable frequency range of the planar response curve, which corresponds to non-planar motions, is moved from DE in Fig. 2 to $x'y'$ in Fig. 4. The reason is that, when $\delta_2 < 0.0$, $h < b$ and the beam is more flexible in the y -direction than in the z -direction. So the motion in the y -direction can be easily autoparametrically excited, and the unstable region of the planar response curve shifts to the left and enlarges. When δ_2 is positive, the unstable region xy shifts to the right and narrows down because the beam is stiffer in the y -direction than in the z -direction. It follows from (16) and (17) that when δ_2 changes the magnitude of the planar response amplitude a_2 does not change because δ_2 appears only in the expression for the coefficient R_3 . It is due to the fact that we have normalized all structure properties with respect to D_η . Figure 5 shows the domain of existence of non-planar responses in the δ_2 - σ plane. As the natural frequency bias δ_2 increases beyond a critical positive value or decreases below a critical negative value, there are no non-planar motions.

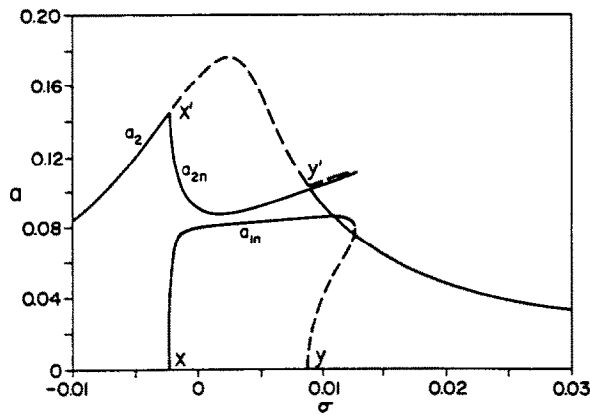


Fig. 4. Response curves of the first mode for a beam with an aspect ratio $b/h \approx 1.0$: mode (1, 1), $\omega_{11} = \omega_{21}$, $\delta_0 = 0.0$, $\delta_2 = -0.01$, $\mu = 0.04$, $\beta_8 f = 0.002$, $\beta_7 = 0.6489$; a_2 = planar response amplitude; a_{1n} , a_{2n} = non-planar response amplitudes; (—) stable, (---) unstable with at least one eigenvalue being positive.

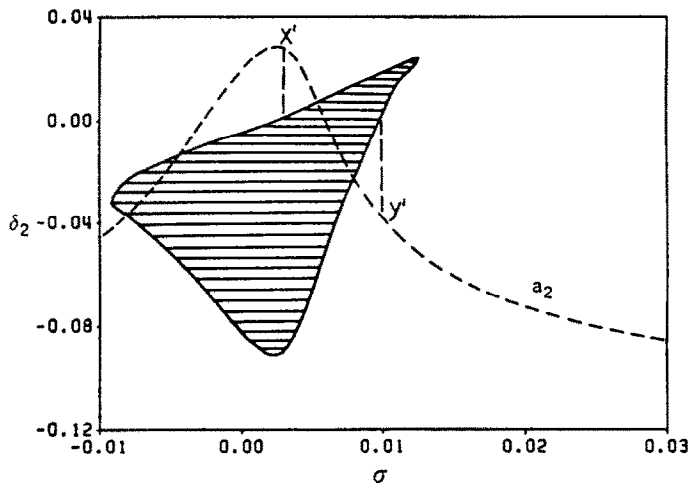


Fig. 5. Variation of the unstable frequency range of planar responses with the natural frequency bias δ_2 : $b/h \approx 1.0$, mode (1, 1), $\omega_{11} = \omega_{21}$, $\delta_0 = 0.0$, $\mu = 0.04$, $\beta_8 f = 0.002$, $\beta_7 = 0.6489$; a_2 = planar response amplitude (for reference).

Figure 6 shows the effect of damping on the unstable region xy of planar motions. Increasing the damping coefficient μ decreases the amplitude a_2 , but it does not change much the region of unstable planar motions. For values of μ around the critical value 0.06, the unstable region is broken into two sections. But checking the non-planar response curves, we find that there is only one non-planar response curve. Increasing the damping coefficient above a critical value stabilizes the planar motion.

If the amplitude of base motion increases to $\beta_8 f = 0.012$, we find that Hopf bifurcations begin to appear in the non-planar response curves. Between these Hopf bifurcation points the unstable fixed-point solutions correspond to amplitude- and phase-modulated motions, which will be investigated later.

To evaluate the influence of non-linear geometric terms, we let $\alpha_i = \beta_i = 0$, $i = 1, 2, 3, 4$, thereby deleting the non-linear geometric terms from (14)–(17). The results show that, without the non-linear geometric terms, the planar response curves of the first mode are bent to the left and are qualitatively as well as quantitatively wrong, as compared with those in Fig. 4. Whereas in the absence of the non-linear geometric terms, non-planar motions can be predicted in the case of external excitations but cannot be predicted in the case of parametric excitations [27], the predicted amplitudes and frequency range are erroneous. These results confirm the fact that the non-linear geometric terms are important for predicting the response of the first mode.

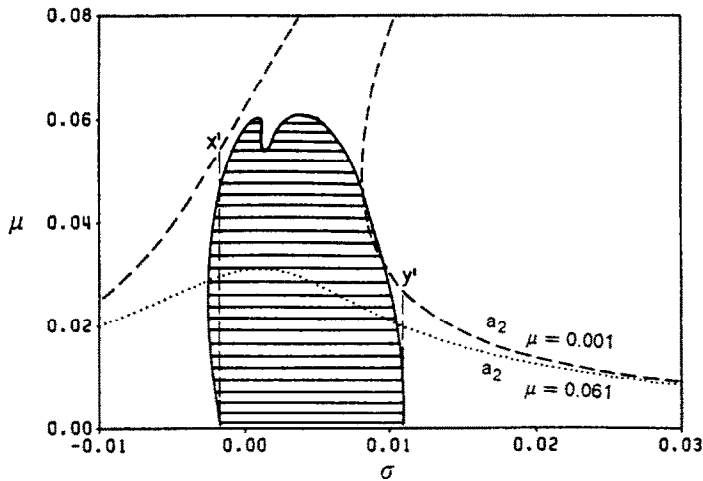


Fig. 6. Variation of the unstable frequency range of planar responses with the damping coefficient μ : $b/h \approx 1.0$, mode (1, 1), $\omega_{11} = \omega_{21}$, $\delta_0 = 0.0$, $\delta_2 = -0.01$, $\beta_8 f = 0.002$, $\beta_7 = 0.6489$; a_2 = planar response amplitude (for reference).

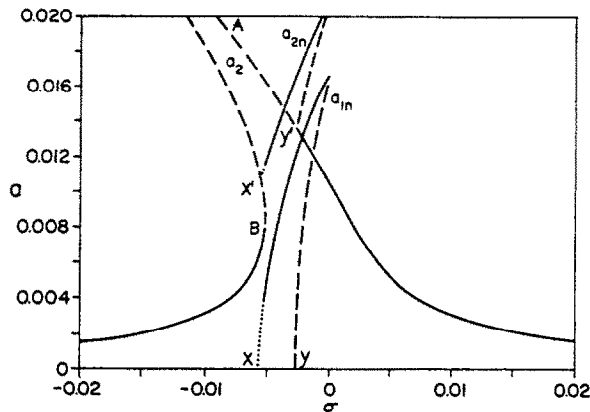


Fig. 7. Response curves of the second mode for a beam with an aspect ratio $b/h \approx 1.0$; mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = -0.02$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$; a_2 = planar response amplitudes; a_{1n} , a_{2n} = non-planar response amplitudes; (—) stable, (---) unstable with at least one eigenvalue being positive, (...) unstable with the real part of a complex conjugate pair of eigenvalues being positive.

Figures 7 and 8 show the response curves of the second mode with δ_2 being equal to -0.02 and 0.002 , respectively. It follows from these figures that the non-planar bifurcation point x' is on the unstable or stable branch of the planar response curve depending on whether $\delta_2 < 0$ or $\delta_2 > 0$, respectively. In Figure 7, the branch Ay' is unstable with respect to out-of-plane disturbances. When δ_2 is close to zero, the unstable region Ay' in Fig. 7 or $x'y'$ in Fig. 8 is large. Furthermore, point B in Fig. 7 is independent of the value of δ_2 , which is different from the results of Haight and King [1]. The effect of the natural frequency bias δ_2 on the unstable frequency range xy of planar motions is shown in Fig. 9. We note that the bifurcation point y' is always on the stable branch of the planar response curve (solid line) and it monotonically shifts to the left as δ_2 decreases. But the bifurcation point x' is on the stable branch when $\delta_2 > 0$ and on the unstable branch (dashed line) when $\delta_2 < 0$. Also we note that x' cannot lie on the stable segment of the left planar response curve (dotted line), which means that the branch AB is always stable irrespective of the value of δ_2 . This character is different from that of the first mode shown in Fig. 2. Furthermore, both Figs 5 and 9 show that the non-planar motions can only exist in a small range of δ_2 if $\delta_2 > 0$. But for $\delta_2 < 0$, bending in the y -direction is more flexible than that in the z -direction. Hence, $v(s, t)$ can be easily autoparametrically excited, and non-planar motions exist in a larger range of δ_2 .

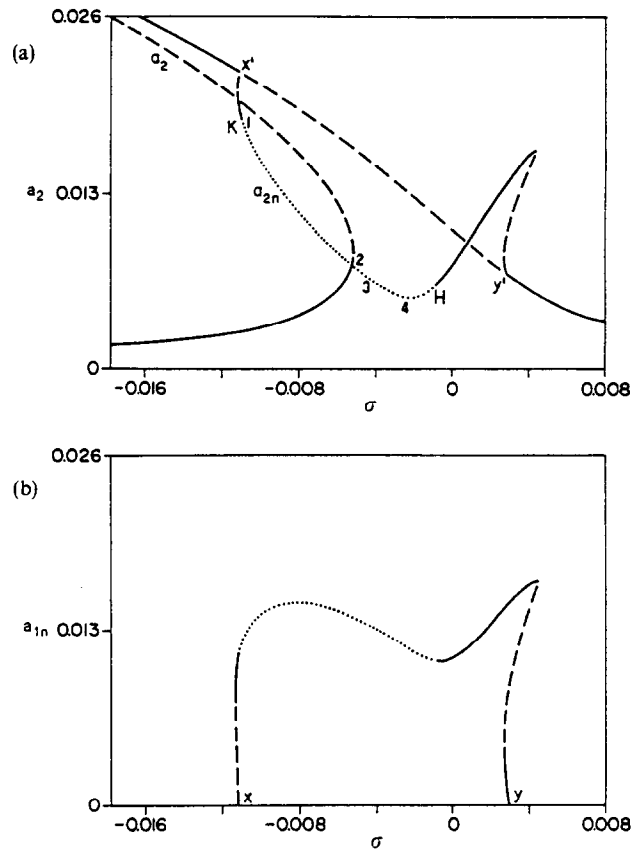


Fig. 8. Response curves of the second mode for a beam with an aspect ratio $b/h \approx 1.0$: mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$; a_2 = planar response amplitudes; a_{1n} , a_{2n} = non-planar response amplitudes; (—) stable, (---) unstable with at least one eigenvalue being positive, (...) unstable with the real part of a complex conjugate pair of eigenvalues being positive; (a) response curves of $w(s, t)$, (b) response curves of $v(s, t)$.

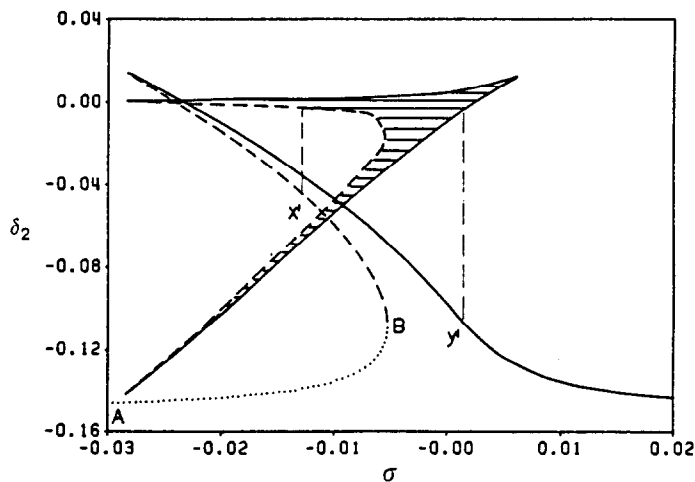


Fig. 9. Variation of the unstable frequency range of planar responses with the natural frequency bias δ_2 : $b/h \approx 1.0$, mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$; a_2 = planar response amplitude (for reference).

Along branch HK of Fig. 8, the fixed-point solutions are unstable with two of the eigenvalues of the Jacobian matrix being complex conjugate having a positive real part. As σ decreases from a value larger than that corresponding to point H, the non-planar fixed point loses stability with a complex conjugate pair of eigenvalues moving into the right-half plane. This corresponds to the extensively studied Hopf bifurcation [28]. Based on the Hopf

bifurcation theorem [29], one expects amplitude- and phase-modulated motions for values of σ near H . To investigate the behavior of the system for values of the parameters corresponding to these unstable fixed-point solutions, we use a Runge–Kutta routine to integrate (22)–(25) for a long period of time to determine the attractors of the modulation equations. Figures 10–13 show the obtained attractors for $\sigma = -0.01117$, -0.005231 , -0.005131 and -0.002 , corresponding to points 1–4 in Fig. 8. The figures in part (b) show that the energy is continuously exchanged between the two modes. Moreover, the phase difference is not equal to 180° as in the case of internal resonance of free vibrations [30]. The points in Figs 10(b), 11(b) and 12(b) denote the unstable fixed-point solutions predicted by the perturbation solution. Figures 10–12 show that changes in the amplitudes is periodic and hence the response of the beam consists of two periods, which are not commensurable in general, and hence the response is a motion on a torus. The spectra of the amplitude a_{2n} are shown in Figs 10(a), 11(a) and 12(a), which confirm that a_{2n} is periodic. We note that when δ_2 is close to zero, the amplitudes seem to be chaotic as shown in Fig. 13. The broadband character of the spectrum of a_{2n} in Fig. 13(a) suggests that the attractor is chaotic. It follows from the harmonic components with the lowest frequency in the spectra in Figs 10–13 that the periods of the wave form of a_{2n} are approximately 250, 300, 167 and

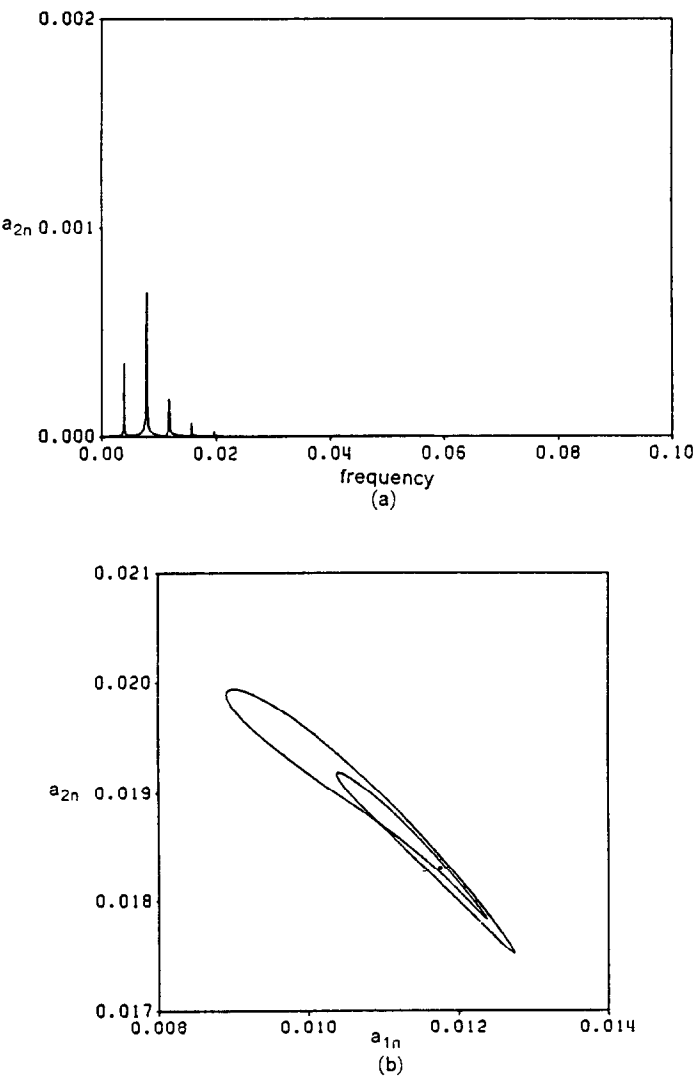


Fig. 10. (a) Fourier harmonic analysis of a_{2n} , (b) a projection of the trajectory onto the a_1 – a_2 plane: $b/h \simeq 1.0$, mode $(2, 2)$, $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$, $\sigma = -0.01117$.

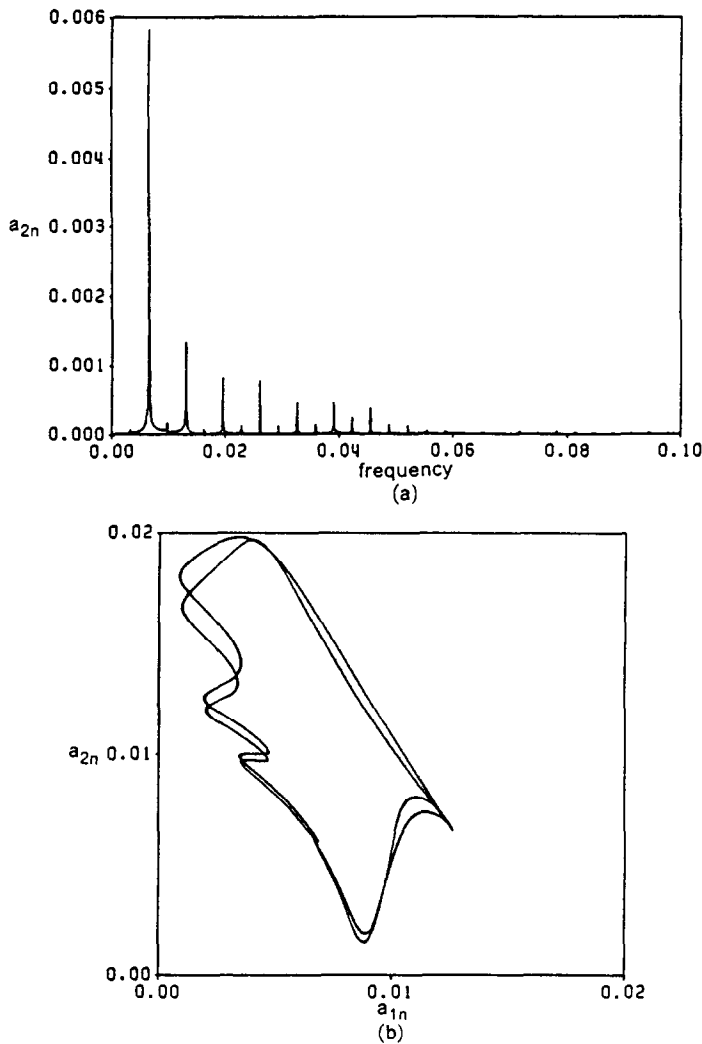


Fig. 11. (a) Fourier harmonic analysis of a_{2n} , (b) a projection of the trajectory onto the a_1 - a_2 plane: $b/h \simeq 1.0$, mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$, $\sigma = -0.005231$.

∞ . Since the linear natural period of the second mode $T = 0.285$, the amplitude-modulated motions shown in Figs 10–12 vary periodically but very slowly. Hence, these variations may be easily overlooked in experiments. Here we want to indicate that, in performing the Fourier harmonic analysis, we did not use any window (or taper) functions, and the zero-frequency components were eliminated in all the figures.

Since the amplitude and phases are not constant but periodic with a period that is larger than that corresponding to free oscillations, the resulting motion is non-periodic having two periods (i.e. motion on a torus). The motion can be better visualized by plotting the motion of the tip-end of the beam (as shown in Fig. 14). Figure 14(a), (b) and (c) shows that the elliptical route keeps changing the lengths of axes and direction, and they also show the twisting motion. But the twisting frequency is very low because the period of the wave amplitude is very large as indicated above. Due to the non-linear terms, the inertia force in the y -direction is not proportional to $v(s, t)$ and the inertia force in the z -direction is not proportional to $w(s, t)$, and hence the resultant inertia force is not parallel to the total displacement in the y - z plane and it induces a twisting moment of the beam. This is a whirling motion of the beating type. Figure 14(d) shows the steady whirling motion for comparison. We note that the tip-end of the beam steadily follows an elliptical route, which is the steady whirling motion. Although the base motion is in the z -direction, a_{1n} is larger than a_{2n} in this case. We know that the linear perfect resonant points are $\sigma = 0$ and

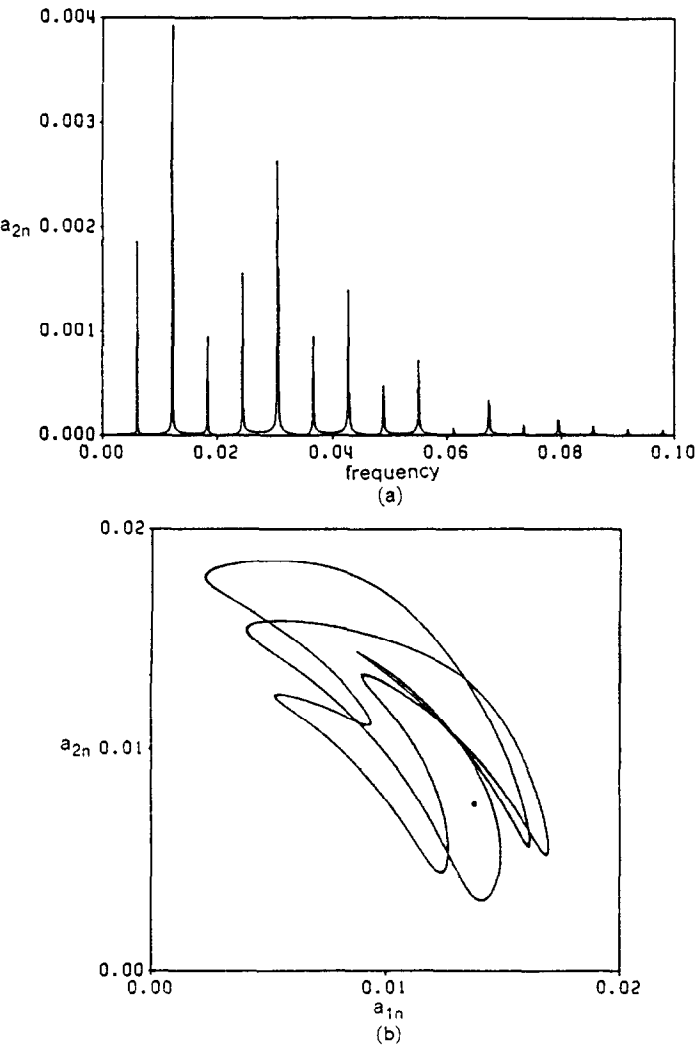


Fig. 12. (a) Fourier harmonic analysis of a_{2n} , (b) a projection of the trajectory onto the a_1 - a_2 plane: $b/h \simeq 1.0$, mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_\gamma = 0.6489$, $\sigma = -0.005131$.

$\delta_2/2(1 + \delta_0)$ for w and v , respectively. But the non-linear softening effects decrease the natural frequency of the second mode, and hence the perfect resonant point of v is moved closer to $\sigma = 0$ than that of w , which results in $a_{1n} > a_{2n}$. It follows from Fig. 14(d) that, for the forced vibration case, the phase difference between w and v is not 90° . In order to “make” the tip-end twist faster for a clearer picture, we put $\varepsilon = 3$ in obtaining Fig. 14. If ε is smaller, the tip-end will twist slowly, the path lines will get closer, and the amplitude will become smaller as can be seen from (19) and (20). But the shape will not change. Another point is that the motion is asymmetric even though the geometry of the structure is symmetric. With different initiations of the twisting motion, the steady twisting motion can exist within the other part of the y - z plane and the route will be the one obtained by rotating the figures shown in Fig. 14 by 180° with respect to the y - or z -axes.

Neglecting the non-linear geometric terms, we find that the response curves of the second mode are similar to those in Fig. 8 but the non-linear geometric terms increase the amplitude of planar responses and narrow down the frequency range of non-planar motions. As expected, the importance of the non-linear geometric terms decreases as the response frequency increases. Moreover, we find that non-planar motions can be unstable even in the absence of non-linear geometric terms, in agreement with Crespo da Silva [3] and in disagreement with Hyer [2].

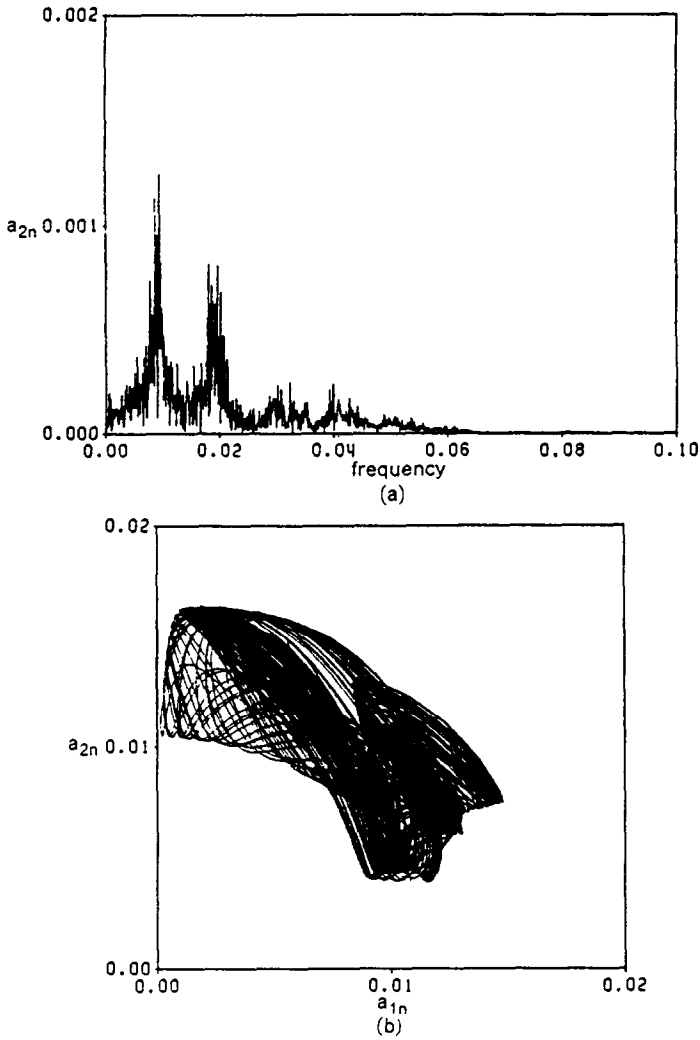


Fig. 13. (a) Fourier harmonic analysis of a_{2n} , (b) a projection of the trajectory onto the a_1 - a_2 plane: $b/h \simeq 1.0$, mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$, $\sigma = -0.002$.

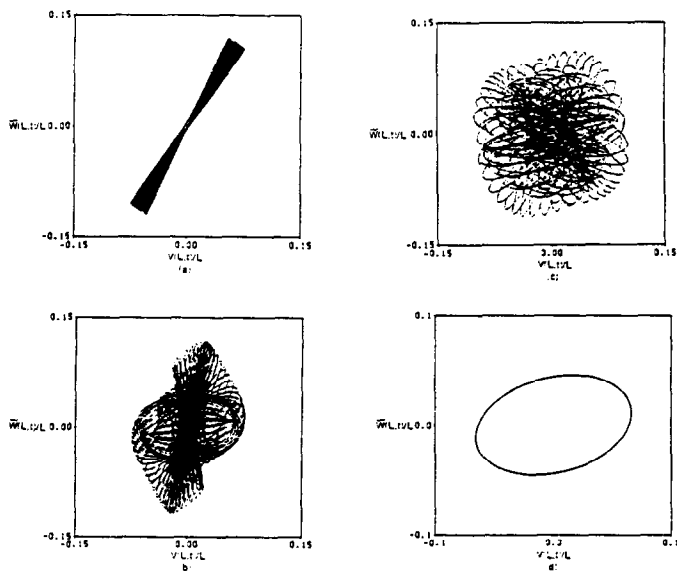


Fig. 14. Path of the tip-end of the beam for the case of an amplitude- and phase-modulated motion: $b/h \simeq 1.0$, mode (2, 2), $\omega_{12} = \omega_{22}$, $\delta_0 = 0.0$, $\delta_2 = 0.002$, $\mu = 0.04$, $\beta_8 f = 0.00006$, $\beta_7 = 0.6489$, $\varepsilon = 3$, (a) $\sigma = -0.01117$, (b) $\sigma = -0.005231$, (c) $\sigma = -0.005131$, (d) $\sigma = 0$ (steady whirling motion).

Case 2: rectangular cross section

In this case, $\delta_0 = -0.9745$ (i.e. $b/h \simeq 6.2673$) and $\beta_y = 0.0352$, and we let $\mu = 0.04$. For this beam, $\omega_{21} = \omega_{12}$. We investigate the possibility of non-planar vibrations consisting of the first mode in the z -direction and the second mode in the y -direction.

When $\delta_2 = 0.02$, Fig. 15 shows that the non-planar response curve consists of two branches, which end at a value of σ far below zero. Except for a small range in which amplitude- and phase-modulated motions exist, the upper branch is unstable. But the lower branch is stable within a very large frequency range below $\sigma = 0$; the amplitudes in both bending directions are of the same order although the flexural rigidity of bending in the z -direction is much larger than that in the y -direction. This phenomenon results from the participation of the non-linear geometric terms due to coupling between torsional and lateral vibrations and the low torsional rigidity of a rectangular compact beam. When $v = 0$, only flexural bending can create a displacement in the z -direction because the torsional motion does not create any displacement in the z -direction. However, the large flexural rigidity in the z -direction resists such bending. When $v \neq 0$, small disturbances in the z -direction can induce large-amplitude vibrations in the z -direction via torsional motion because the torsional rigidity is low. This also explains why the route of the tip-end of the

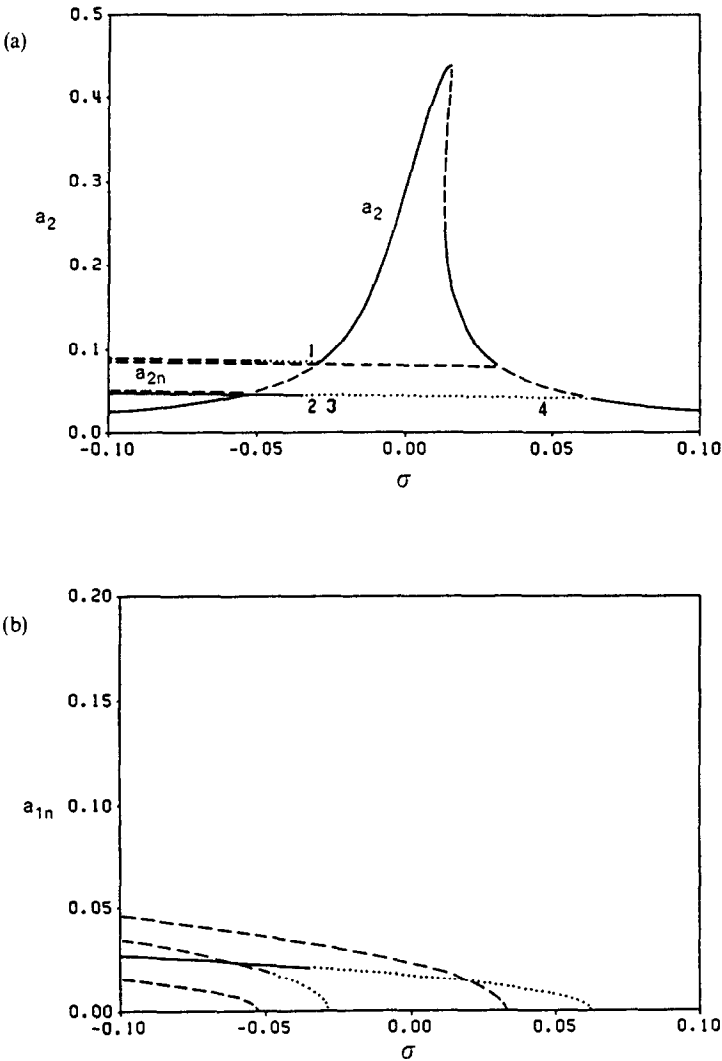


Fig. 15. Response curves of a beam with an aspect ratio $b/h \simeq 6.2673$: mode (2, 1), $\omega_{12} = \omega_{21}$, $\delta_0 = -0.9745$, $\delta_2 = 0.02$, $\mu = 0.04$, $\beta_8 f = 0.005$, $\beta_y = 0.0352$; a_2 = planar response amplitude; a_{1n} , a_{2n} = non-planar response amplitudes; (—) stable, (---) unstable with at least one eigenvalue being positive, (...) unstable with the real part of a complex conjugate pair of eigenvalues being positive; (a) response curves of $w(s, t)$, (b) response curves of $v(s, t)$.

beam is a long ellipse (as shown in Fig. 16). The gap inside the curve is due to the base motion in z -direction. It follows from the positions of the cross section indicated in this figure that the frequency of the torsional motion is equal to 2Ω . Since we neglected the torsional inertia, the rotation angle here is due to the twisting moment induced by inertia forces. Hence, the torsional inertia may not be negligible for a rectangular beam.

When $\delta_2 = -0.001$, the response curves are shown in Fig. 17. We note that there is only one unstable region in the planar response curves due to out-of-plane disturbances. Checking four fixed-point solutions, points 1–4 in Fig. 15, we find that all of these unstable fixed-point solutions having a pair of complex conjugate eigenvalues with positive real parts correspond to chaotic motions.

Neglecting the non-linear geometric terms, we find that the frequency range of non-planar motions is different and the bending directions of non-planar response curves are opposite. It confirms again that the non-linear geometric terms are not negligible for low-frequency responses.

In the present analysis, we investigated the dynamic behavior of out-of-plane, one-to-one modal interactions. But, here we want to indicate that three-to-one in-plane modal coupling between the second and third modes may occur in this system because $\omega_{i3}/\omega_{i2} \approx 3$ and our equations contain cubic non-linearities [5, 27]. Furthermore, if torsional damping is not zero, the twisting vibration of non-planar motion may be suppressed or damped out. For rectangular beams, the torsional inertia may have an influence on the responses because the non-planar motion is induced by torsional motions.

5. CONCLUSIONS

A combination of the method of multiple scales and a Galerkin procedure is used to analyse the non-linear response of a cantilever beam subject to lateral harmonic base excitations. The analysis focuses on the case of a one-to-one internal (autoparametric) resonance in which the linear natural frequency of flexural vibrations in one plane is approximately equal to the linear natural frequency of flexural vibrations in the other plane. Four first-order ordinary-differential equations are derived for the modulation of the amplitudes and phases of the two interacting modes with damping, non-linearity, auto-parametric resonance, and primary external resonance. The analysis takes into account both the inertia and geometric non-linearities. The modulation equations are used to determine the fixed points and their stability.

The results show that the inertia non-linearity is of the softening type, whereas the geometric non-linearity is of the hardening type. Consequently, the overall effective

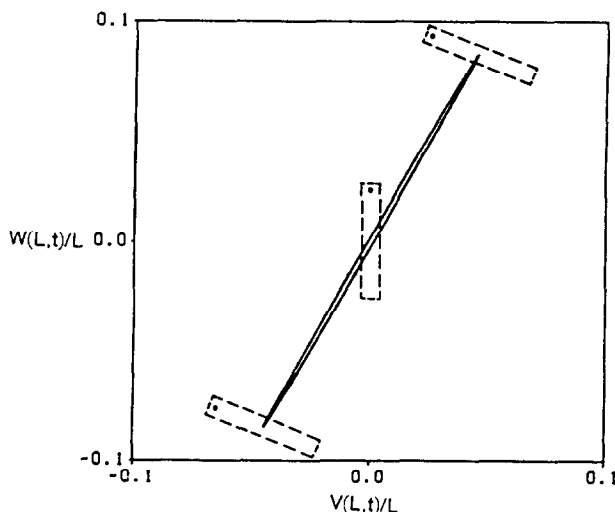


Fig. 16. The path of the tip-end of the beam for the case of a steady whirling motion: $b/h \approx 6.2673$, mode (2, 1), $\omega_{12} = \omega_{21}$, $\delta_0 = -0.9745$, $\delta_2 = 0.02$, $\mu = 0.04$, $\beta_8 f = 0.005$, $\beta_7 = 0.0352$, $\varepsilon = 1$, $\sigma = -0.05$.

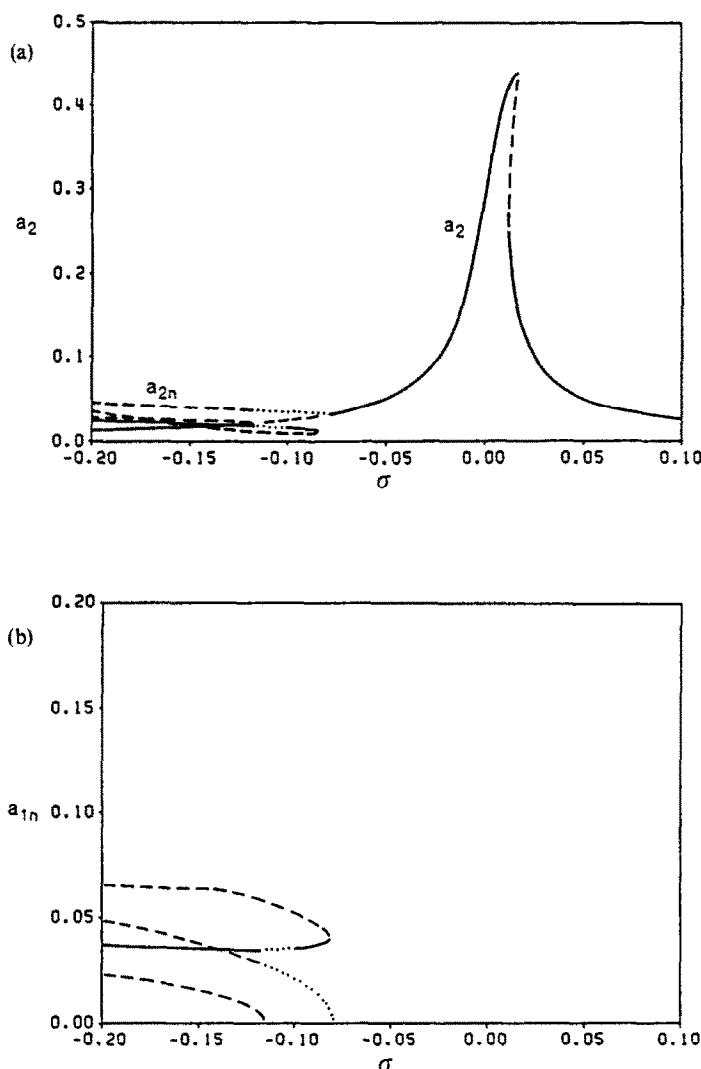


Fig. 17. Response curves of a beam with an aspect ratio $b/h \approx 6.2673$: mode (2, 1), $\omega_{12} = \omega_{21}$, $\delta_0 = 0.9745$, $\delta_2 = -0.001$, $\mu = 0.04$, $\beta_3 f = 0.005$, $\beta_7 = 0.0352$; a_2 = planar response amplitude; a_{1n}, a_{2n} = non-planar response amplitudes; (—) stable, (---) unstable with at least one eigenvalue being positive, (...) unstable with the real part of a complex conjugate pair of eigenvalues being positive; (a) response curves of $w(s, t)$, (b) response curves of $v(s, t)$.

non-linearity may be hardening or softening depending on the order of the mode. For the first mode, the effective non-linearity is of the hardening type, whereas for the second and higher modes it is of the softening type. Therefore, neglecting the geometric non-linearity leads to qualitatively erroneous frequency–response curves for the first mode and underpredicts the planar response amplitudes of the second and higher modes. The error decreases as the mode number increases. These conclusions hold for planar motions in which only one mode is excited. Furthermore, the natural frequency bias δ_2 controls shifting of the unstable frequency range of the planar response curve and hence the development of an interval of non-planar motions. Damping can suppress and stabilize planar responses. By changing the natural frequency bias and damping coefficient, one can suppress non-planar motions.

The results show that planar motions in the z -direction may be destabilized by disturbances in the y -direction, resulting in vibrations in both planes. For certain beam and load parameters, non-planar responses can be (a) steady whirling motions (the principal axes of the elliptical route are not parallel to the y - or z -axes), (b) unsteady whirling motions (lengths and directions of axes of the elliptical route continuously change and the motion also shows a low-frequency twisting motion), or (c) chaotic motions. The unsteady whirling motions may be periodic if the period of the amplitudes is commensurable with that of the base motion. But this periodicity is not obvious because the twisting frequency is too low.

For a rectangular beam, steady whirling motions exist in a large frequency range below $\sigma = 0$ because the torsional rigidity of a rectangular compact beam is very small. However, the torsional inertia may not be negligible for rectangular beams. Around the resonant point $\sigma = 0$, most of the non-planar motions correspond to chaotic motions for the cases we investigated.

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REFERENCES

1. E. C. Haight and W. W. King, Stability of non-linear oscillations of an elastic rod. *J. Acoust. Soc. Am.* **52**, 899–911 (1972).
2. M. W. Hyer, Whirling of a base-excited cantilever beam. *J. Acoust. Soc. Am.* **65**, 931–939 (1979).
3. M. R. M. Crespo da Silva, On the whirling of a base-excited cantilever beam. *J. Acoust. Soc. Am.* **67**, 704–707 (1980).
4. C.-H. Ho, R. A. Scott and J. G. Easley, Non-planar, non-linear oscillations of a beam—I. Forced motions. *Int. J. Non-Linear Mech.* **10**, 113–127 (1975).
5. A. H. Nayfeh and D. T. Mook, *Nonlinear Oscillations*. Wiley-Interscience, New York (1979).
6. A. H. Nayfeh, Nonlinear transverse vibrations of beams with properties that vary along the length. *J. Acoust. Soc. Am.* **53**, 766–770 (1973).
7. E. G. Tezak, D. T. Mook and A. H. Nayfeh, Nonlinear analysis of the lateral response of columns to periodic loads. *J. Mech. Des.* **100**, 651–659 (1978).
8. A. H. Nayfeh, D. T. Mook and S. Sridhar, Nonlinear analysis of the forced response of structural elements. *J. Acoust. Soc. Am.* **55**, 281–291 (1974).
9. J. F. Nayfeh, A. H. Nayfeh and D. T. Mook, Modal interactions in the response of beams to a harmonic excitation. The Second Conference on Nonlinear Vibrations, Stability, and Dynamics of Structures and Mechanisms, Blacksburg, Virginia (1–3 June 1988).
10. D. H. Hodges and E. H. Dowell, Nonlinear equations of motion for the elastic bending and torsion of twisted nonuniform rotor blades. NASA TN D-7818 (1974).
11. E. H. Dowell, J. Traybar and D. H. Hodges, An experimental-theoretical correlation study of nonlinear bending and torsion deformations of a cantilever beam. *J. Sound Vib.* **50**, 533–544 (1977).
12. A. Rosen and P. Friedmann, The nonlinear behavior of elastic slender straight beams undergoing small strains and moderate rotations. *J. appl. Mech.* **46**, 161–168 (1979).
13. W. K. Tso, Parametric torsional stability of a bar under axial excitation. *J. appl. Mech.* **35**, 13–19 (March 1968).
14. K. Sato, H. Saito and K. Otomi, The parametric response of a horizontal beam carrying a concentrated mass under gravity. *J. appl. Mech.* **45**, 643–648 (1978).
15. V. V. Bolotin, *The Dynamic Stability of Elastic Systems* (Translated by V. I. Weingarten *et al.*). Holden-Day, San Francisco (1964).
16. M. L. Moody, The parametric response of imperfect columns, in *Developments in Mechanics*, Vol. 4, *Proc. Tenth Midwestern Mechanics Conference*, pp. 329–346 (1967).
17. M. R. M. Crespo da Silva and C. C. Glynn, Nonlinear flexural–flexural–torsional dynamics of inextensional beams—I. Equations of motion. *J. Struct. Mech.* **6**, 437–448 (1978).
18. M. R. M. Crespo da Silva and C. C. Glynn, Nonlinear flexural–flexural–torsional dynamics of inextensional beams—II. Forced motions. *J. Struct. Mech.* **6**, 449–461 (1978).
19. R. M. Evan-Iwanowski, W. F. Sanford and T. Kehagioglou, Nonstationary parametric response of a nonlinear column. *Dev. theor. appl. Mech.* **5**, 715–743 (1970).
20. M. R. M. Crespo da Silva and D. H. Hodges, Nonlinear flexure and torsion of rotating beams, with application to helicopter rotor blades—I. Formulation. *Vertica* **10**, 151–169 (1986).
21. M. R. M. Crespo da Silva and D. H. Hodges, Nonlinear flexure and torsion of rotating beams, with application to helicopter rotor blades—II. Response and stability results. *Vertica* **10**, 171–186 (1986).
22. T. D. Burton and M. Kolowith, Nonlinear resonances and chaotic motion in a flexible parametrically excited beam. The Second Conference on Nonlinear Vibrations, Stability, and Dynamics of Structures and Mechanisms, Blacksburg, Virginia (1–3 June 1988).
23. A. G. Haddow and S. M. Hasan, Nonlinear oscillation of a flexible cantilever: experimental results. The Second Conference on Nonlinear Vibrations, Stability, and Dynamics of Structures and Mechanisms, Blacksburg, Virginia (1–3 June 1988).
24. A. H. Nayfeh, *Introduction to Perturbation Techniques*. Wiley-Interscience, New York (1981).
25. C. F. Gerald, *Applied Numerical Analysis*, 2nd edn, pp. 8–10. Addison-Wesley, Reading, Massachusetts (1983).
26. S. P. Timoshenko and J. N. Goodier, *Theory of Elasticity*, 3rd edn, pp. 312–313. McGraw-Hill, New York (1970).
27. A. H. Nayfeh and P. F. Pai, Non-linear non-planar parametric responses of an inextensional beam. *Int. J. Non-Linear Mech.* **24**, 139–158 (1989).
28. J. E. Marsden and M. McCracken, *Hopf Bifurcation and Its Application*. Springer, New York (1976).
29. J. K. Hale, *Oscillations in Nonlinear Systems*. McGraw-Hill, New York (1963).
30. M. R. M. Crespo da Silva and C. C. Glynn, Non-linear non-planar resonant oscillations in fixed-free beams with support asymmetry. *Int. J. Solids Struct.* **15**, 209–219 (1979).

APPENDIX A

$$\begin{aligned}
\beta_1 &= \int_0^1 \Phi_n \left[\Phi_m'' \int_1^s \Phi_m'' \Phi_n'' ds - \Phi_m'' \int_0^s \Phi_n'' \Phi_m' ds \right]' ds \\
\beta_2 &= \int_0^1 \Phi_n \left[\Phi_m'' \int_0^s \int_1^s \Phi_m'' \Phi_n'' ds ds \right]'' ds \\
\beta_3 &= \int_0^1 \Phi_n [\Phi_n' (\Phi_m' \Phi_m'')] ds \\
\beta_4 &= \int_0^1 \Phi_n [\Phi_n' (\Phi_n' \Phi_n'')] ds \\
\beta_5 &= \int_0^1 \Phi_n \left[\Phi_n' \int_1^s \int_0^s \Phi_n'^2 ds ds \right]' ds \\
\beta_6 &= \int_0^1 \Phi_n \left[\Phi_n' \int_1^s \int_0^s \Phi_m'^2 ds ds \right]' ds \\
\beta_7 &= \int_0^1 [(s-1)\Phi_n \Phi_n'' + \Phi_n \Phi_n'] ds \\
\beta_8 &= \int_0^1 \Phi_n ds
\end{aligned}$$

α_i for mode $(m, n) = \beta_i$ for mode (n, m) .

APPENDIX B

$$\begin{aligned}
R_1 &= \mu \omega_{1m} \\
R_2 &= \frac{1}{4} \left[-2\alpha_6 \omega_{2n}^2 + \delta_0 \alpha_1 + \frac{\delta_0^2}{\beta_\gamma} \alpha_2 + (1 + \delta_0) \alpha_3 \right] \\
R_3 &= 2\omega_{1m} \omega_{2n} \dot{\sigma} - \delta_2 z^4 + \alpha_7 g_0 \\
R_4 &= \frac{1}{2} \alpha_5 \omega_{1m}^2 - \frac{3}{4} (1 + \delta_0) \alpha_4 \\
R_5 &= \frac{1}{2} \left(\delta_0 \alpha_1 + \frac{\delta_0^2}{\beta_\gamma} \alpha_2 + (1 + \delta_0) \alpha_3 \right) \\
E_1 &= \mu \omega_{2n} \\
E_2 &= \frac{1}{4} \left(2\beta_6 \omega_{1m}^2 + \delta_0 \beta_1 - \frac{\delta_0^2}{\beta_\gamma} \beta_2 - \beta_3 \right) \\
E_3 &= 2\omega_{2n}^2 \dot{\sigma} + \beta_7 g_0 \\
E_4 &= \frac{1}{2} \beta_5 \omega_{2n}^2 - \frac{3}{4} \beta_4 \\
E_5 &= \frac{1}{2} \left(\delta_0 \beta_1 - \frac{\delta_0^2}{\beta_\gamma} \beta_2 - \beta_3 \right).
\end{aligned}$$